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Systems and methods for percutaneous intravascular access and guidewire placement

Abstract

A method of creating intravascular access includes positioning the main body of a device within a primary vessel and manipulating a guidewire having a second alignment member to a location proximate to an inner wall of a secondary vessel. Engaging the alignment member of the distal end of the device and the second alignment member of the guidewire through the respective walls of the primary and secondary vessels, so that the device and guidewire are in close proximity and in alignment, thereby pushing the primary and secondary vessels together. Extending the piercing member distally from the main body, through the wall of the primary vessel, and through an adjacent wall of the secondary vessel, so that the end of the piercing member is disposed within the secondary vessel and thereby creating a communicating aperture on the opposing walls of the primary and secondary vessel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional application under 35 U.S.C. 120 of commonly assigned U.S. patent application Ser. No. 15/808,711, entitled Systems and Methods for Percutaneous Intravascular Access and Guidewire Placement, filed on Nov. 9, 2017, which claims the benefit under 35 U.S.C. 119(e) of the filing date of Provisional U.S. Application Ser. No. 62/421,125, entitled Systems and Methods for Percutaneous Intravascular Access and Guidewire Placement, filed on Nov. 11, 2016, which application is expressly incorporated herein by reference, in its entirety.

BACKGROUND OF THE INVENTION

(1) In the body, various fluids are transported through conduits throughout the organism to perform various essential functions. Blood vessels, arteries, veins, and capillaries carry blood throughout the body, carrying nutrients and waste products to different organs and tissues for processing. Bile ducts carry bile from the liver to the duodenum. Ureters carry urine from the kidneys to the bladder. The intestines carry nutrients and waste products from the mouth to the anus.

(2) In medical practice, there is often a need to connect conduits to one another or to a replacement conduit to treat disease or dysfunction of the existing conduits. The connection created between conduits is called an anastomosis.

(3) In blood vessels, anastomoses are made between veins and arteries, arteries and arteries, or veins and veins. The purpose of these connections is to create either a high flow connection, or fistula, between an artery and a vein, or to carry blood around an obstruction in a replacement conduit, or bypass. The conduit for a bypass is a vein, artery, or prosthetic graft.

(4) An anastomosis is created during surgery by bringing two vessels or a conduit into direct contact. The vessels are joined together with suture or clips. The anastomosis can be end-to-end, end-to-side, or side-to-side. In blood vessels, the anastomosis is elliptical in shape and is most commonly sewn by hand with a continuous suture. Other methods for anastomosis creation have been used including carbon dioxide laser, and a number of methods using various connecting prosthesis, clips, and stents.

(5) An arterio-venous fistula (AVF) is created by connecting an artery to a vein. This type of connection is used for hemodialysis, to increase exercise tolerance, to keep an artery or vein open, or to provide reliable access for chemotherapy.

(6) An alternative is to connect a prosthetic graft from an artery to a vein for the same purpose of creating a high flow connection between artery and vein. This is called an arterio-venous graft, and requires two anastomoses. One is between artery and graft, and the second is between graft and vein.

(7) A bypass is similar to an arteriovenous graft. To bypass an obstruction, two anastomoses and a conduit are required. A proximal anastomosis is created from a blood vessel to a conduit. The conduit extends around the obstruction, and a second distal anastomosis is created between the conduit and vessel beyond the obstruction.

(8) As noted above, in current medical practice, it is desirable to connect arteries to veins to create a fistula for the purpose of hemodialysis. The process of hemodialysis requires the removal of blood from the body at a rapid rate, passing the blood through a dialysis machine, and returning the blood to the body. The access to the blood circulation is achieved with (1) catheters placed in large veins, (2) prosthetic grafts attached to an artery and a vein, or (3) a fistula where an artery is attached directly to the vein.

(9) Hemodialysis is required by patients with kidney failure. A fistula using native blood vessels is one way to create high blood flow. The fistula provides a high flow of blood that can be withdrawn from the body into a dialysis machine to remove waste products and then returned to the body. The blood is withdrawn through a large access needle near the artery and returned to the fistula through a second large return needle. These fistulas are typically created in the forearm, upper arm, less frequently in the thigh, and in rare cases, elsewhere in the body. It is important that the fistula be

able to achieve a flow rate of 500 ml per minute or greater, in order for the vein to mature or grow. The vein is considered mature once it reaches >4 mm and can be accessed with a large needle. The segment of vein in which the fistula is created needs to be long enough (>6 cm) to allow adequate separation of the access and return needle to prevent recirculation of dialyzed and non-dialyzed blood between the needles inserted in the fistula.

(10) Fistulas are created in anesthetized patients by carefully dissecting an artery and vein from their surrounding tissue, and sewing the vessels together with fine suture or clips. The connection thus created is an anastomosis. It is highly desirable to be able to make the anastomosis quickly, reliably, with less dissection, and with less pain. It is important that the anastomosis is the correct size, is smooth, and that the artery and vein are not twisted.

SUMMARY OF THE INVENTION

(11) The present disclosed invention eliminates the above described open procedures, reduces operating time, and allows for a consistent and repeatable fistula creation.

(12) The present invention comprises a device to allow passage of a guidewire from a primary blood vessel to an adjacent secondary blood vessel, which comprises a main body having a primary lumen and a secondary lumen and a piercing member disposed in the secondary lumen, and configured to be moved distally out of the secondary lumen, and to pierce through tissue while being distally moved. A third lumen located within the piercing member is configured to allow placement of a guidewire from the primary blood vessel to the adjacent secondary blood vessel.

(13) In one embodiment, the secondary lumen is constructed out of superelastic material, such as Nitinol, that is shaped such that the distal tip is oriented toward the adjacent secondary blood vessel. The secondary lumen may have a “J” shape heat set into the secondary lumen; however, different shapes may be used depending upon the type of anatomy that is being accessed. The primary lumen is configured with a stiffness such that it has the ability to straighten the shape of the secondary lumen. Either advancing or retracting the primary lumen relative to the secondary lumen can adjust the rise, or shape, of the secondary lumen. Shaping the primary lumen can further modify the angle at which the piercing member exits the secondary lumen. In an alternative embodiment, the shape of the secondary lumen may be modified using a tendon wire. In still another embodiment, the piercing member is designed to remain in a substantially straight configuration.

(14) In another aspect of the invention, the distal tip of the secondary lumen has a feature to make it such that it will not perforate the blood vessel as it is being placed into a desired position within the body. In the first embodiment noted above, the tip has a large diameter polymer tip that has a rounded distal edge and is atraumatic. This distal tip also has features that make it visible under different imaging techniques, such as ultrasound, fluoroscopy, CT, or MRI. There is a coil constructed of a radiopaque material, embedded in the polymer tip. Small particles of air or other radiopaque materials known to those skilled in the art can also be used to increase the radiopacity of the tip.

(15) In another embodiment of the invention, the feature located on the distal tip of the secondary lumen actively assists in the positioning of the tip relative to the adjacent secondary blood vessel. The tip positioning assistance feature can be accomplished using magnets, sensors, ultrasound or combination of thereof. In one aspect of the invention the feature is composed of a ring magnet that surrounds the secondary lumen and is located near the distal end. A secondary magnet, oriented such that it attracts to the first magnet, is placed in the secondary blood vessel at the desired puncture location. As the secondary lumen, with the cylindrical magnet located on distal end, is advanced in the first blood vessel and comes into proximity of the secondary magnet they will attract and be drawn together. Preferably the secondary magnet is attached to the flexible distal end of a guidewire which allows the secondary magnet to align to the primary magnet, although it could be located on a sheath, balloon catheter, or similar elongated structure. Preferably the magnets are constructed from Neodymium ND-52 to achieve high coercive forces in the smallest

form factor; however other grades of magnets and materials may be used to achieve the desired functionality.

(16) In another embodiment the feature located on the distal tip is a magnetic field sensor which detects a magnet that is placed in the secondary blood vessel at the desired piercing location. One such sensor can be a Lorentz force based MEMS magnetic field sensor, although other proximity sensor types, such as inductive, Hall effect, Doppler effect, or capacitive may be used. The readout of the sensor may be a graphical representation of the magnetic field strength or an audible tone. The distal tip is manipulated until the point at which the magnetic field is strongest, which indicates that the distal tip is aligned with the magnet located in the secondary blood vessel.

(17) In yet another embodiment the feature located on the distal tip of the secondary lumen may have an ultrasound transducer located within it. The ultrasound transducer provides the user with a forward facing ultrasound image of the vasculature. This image assists the user in steering or guiding the device within the vasculature to the desired location. In addition, the ultrasound guidance is used to visualize the adjacent secondary blood vessel. Visualization of the secondary blood vessel allows the user to center the distal tip of the secondary lumen on the blood vessel prior to advancing the hollow piercing element. If the secondary blood vessel is an artery, a continuous wave Doppler ultrasonograph could be used to produce an audible tone to indicate the proximity to the artery. The user would manipulate the position the distal end of the secondary lumen until the tone is most prevalent prior to advancing the hollow piercing element.

(18) The hollow piercing member has a sharp point on the distal tip that exits from the primary vessel by puncturing its wall and enters into the secondary vessel in the same manner. In one embodiment, the sharp distal point is constructed using a lancet point. The primary bevel is ground at an angle between 12 and 20 degrees with a secondary angle between 5-20 degrees, with a rotation angle between 25-45 degrees. The needle grind is designed such that it pierces through the vessel wall and does not core, or cut a plug, through the vessel wall, to minimize bleeding between vessels when removed after the guidewire is placed into the secondary vessel. The outer diameter of the piercing member is also minimized to further reduce bleeding. The piercing member is oriented within the secondary lumen such that the tip of the lancet point is directed toward the adjacent secondary vessel. Other piercing mechanisms, or needle point grind configurations, known to those skilled in the art may be provided.

(19) More particularly, there is provided a device for creating intravascular access and guidewire placement, which comprises a main body having a first lumen, a piercing member disposed in that lumen, and configured to be moved distally out of said lumen and to pierce through tissue while being distally moved, and a handle attached to the main body and having an actuator for moving the piercing member. A second lumen is disposed within the piercing member. A guidewire is disposed in the second lumen for delivery into a desired site from a distal end of the second lumen. The piercing member has a sharp point on one end thereof.

(20) In one disclosed embodiment a ring magnet is attached to the distal tip of the primary lumen which is placed in a primary blood vessel and a guidewire with magnetic tip is placed in the secondary blood vessel. The two magnets are polarized so they attract when they come into proximity with each other. A hollow piercing element is disposed in the primary lumen, and is configured to be moved distally out of said lumen. As the piercing element is moved distally it is configured such that it pierces the tissue between the blood vessels. As the piercing element enters the second blood vessel it will contact the opposing magnet located and disconnect it from the ring magnet. The movement of the opposing magnet provides visual feedback to the user that the piercing element has entered the secondary blood vessel. A guidewire is disposed in the lumen of the piercing element for delivery into the secondary lumen.

(21) In another disclosed embodiment, a third lumen is disposed within the main body, outwardly of the first lumen. The piercing member is retractable into the first lumen. The third lumen is defined by a needle guide having shape memory properties, the needle guide being actuable to a

curved orientation by adjustment of a position of the main body to create an incrementally adjustable radius of curvature on the needle guide. The piercing member has shape memory properties, and is actuatable to create an incrementally adjustable radius of curvature.

(22) The actuator for moving the piercing needle linearly comprises a slide. In the curved embodiment, a second actuator is disposed on the handle for actuating the needle guide to a curved orientation. This actuator comprises a rotatable knob. In both embodiments, the first lumen is defined by a needle guide having an atraumatic distal tip having a relatively large diameter. The atraumatic distal tip can be comprised of a polymer material and further comprises radiopaque materials. In other Preferably, the radiopaque materials comprise a plurality of coils constructed of a radiopaque material.

(23) The sharp point preferably comprises a lancet point and primary bevels.

(24) In another aspect of the invention, there is disclosed a method of creating intravascular access and guidewire delivery, which comprises steps of positioning the main body of a device within a primary vessel and manipulating a distal end of the device to engage an inner wall of the primary vessel and to push the primary vessel into close engagement with an adjacent secondary vessel. Yet another step comprises extending the piercing member distally from the main body, through the wall of the primary vessel, and through an adjacent wall of the secondary vessel, so that the end of the piercing member is disposed within the secondary vessel for creating communicating aperture on the opposing walls of the primary and secondary vessel.

(25) In one embodiment, the method comprises a further step of incrementally adjusting a radius of curvature of the piercing member. In both embodiments, the positioning step is performed percutaneously.

(26) The method further comprises a step of advancing a guidewire distally through a lumen in the piercing member from the primary vessel into the secondary vessel, and a step of withdrawing the device from the vessel, thus leaving the guidewire in place and crossing from the primary vessel to the secondary vessel through said communicating aperture.

(27) In still another aspect of the invention, a method of creating a passage between adjacent primary and secondary blood vessels is disclosed, comprising a step of positioning a main body of the device within the primary vessel and extending a piercing member distally from the main body, through the wall of the primary vessel, and through an adjacent wall of the secondary vessel, so that the piercing member is disposed within the secondary vessel. The secondary lumen is linearly actuated to move relative to a distal end of the piercing member for articulating the distal end of the piercing member for cutting a small communicating aperture from the primary blood vessel to the adjacent secondary blood vessel.

(28) The method further comprises the step of advancing a guidewire distally within the piercing element to pass from the primary blood vessel, while maintaining position substantially within the primary blood vessel, to the adjacent secondary blood vessel.

(29) In another aspect of the invention, there is disclosed a device for creating intravascular access and guidewire placement. The device comprises a main body having a lumen, a piercing member, having a sharp point on one end thereof, disposed in the lumen, and is configured to be moved distally out of the lumen and to pierce through tissue while being distally moved. A handle is attached to the main body and has an actuator for moving the piercing member. A needle guide is provided for guiding the piercing member, the needle guide having a distal end which comprises a first alignment member. Additionally, there is provided a guidewire having a distal tip with a second alignment member disposed on the guidewire distal tip. The piercing member may be retractable into the main body lumen.

(30) The actuator may comprise a rotatable knob, and the sharp point of the piercing member may comprise a lancet point and primary bevels. The first alignment member may comprise a magnetic attachment, at least one magnetic implant, a proximity sensor, an ultrasonic sensor, or other suitable system for alignment of devices disposed on opposing sides of opaque tissue. Similarly, the second

alignment member may comprise a magnetic attachment or implant, a proximity sensor, or an ultrasonic sensor, as well as any other suitable system as discussed above.

(31) In still another aspect of the invention, there is described a method of creating intravascular access, which comprises steps of positioning the main body of a device within a primary vessel, manipulating a distal end of the device having an alignment member to a location proximate to an inner wall of the primary vessel, and manipulating a guidewire having a second alignment member to a location proximate to an inner wall of a secondary vessel. Additional steps include engaging the alignment member of the distal end of the device and the second alignment member of the guidewire through the respective walls of the primary and secondary vessels, so that the device and guidewire are in close proximity and in alignment, and thereby pushing the primary and secondary vessels together, and extending a piercing member distally from the main body, through the wall of the primary vessel, and through an adjacent wall of the secondary vessel, so that the end of the piercing member is disposed within the secondary vessel and thereby creating a communicating aperture on the opposing walls of the primary and secondary vessel. Then, a guidewire is advanced through the lumen of the piercing element into the secondary vessel.

(32) A further step of the inventive method is one of contacting the second alignment member with the needle to dislodge and advance both the second alignment member and the attached guidewire. The positioning step may be performed percutaneously. A further step comprises withdrawing the device from the vessel. Notably, the alignment member on the device may be comprised of one or more of a magnetic attachment, a magnetic implant, a proximity sensor, and an ultrasonic sensor. Similarly, the same is true of the second alignment member.

(33) The invention, together with additional features and advantages thereof, may best be understood by reference to the following description taken in conjunction with the accompanying illustrative drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1a is a view of one embodiment of the device of the present invention, wherein the device has been percutaneously or surgically positioned at a desired location in a blood vessel;
- (2) FIG. 1b is a view, similar to FIG. 1a, of another embodiment of the device of the present invention, wherein the device has been percutaneously or surgically positioned at a desired location in a blood vessel;
- (3) FIG. 2a is a view of the FIG. 1a embodiment of the present invention, illustrating the distal piercing element in isolation;
- (4) FIG. 2b is a view, similar to FIG. 2a, of the embodiment of FIG. 1b, illustrating the distal piercing element in isolation;
- (5) FIG. 3a is a view similar to FIG. 2a, wherein the distal piercing element of FIG. 2a has been advanced distally to push the blood vessel in which it is disposed into contact with the adjacent blood vessel;
- (6) FIG. 3b is a view similar to FIG. 2b, wherein the distal piercing element of FIG. 2b has been advanced distally to push the blood vessel in which it is disposed into contact with the adjacent blood vessel;
- (7) FIG. 4a is a view similar to FIG. 3a, wherein the piercing element is advanced from the primary blood vessel into the adjacent secondary blood vessel;
- (8) FIG. 4b is a view similar to FIG. 3b, wherein the piercing element is advanced from the primary blood vessel into the adjacent secondary blood vessel;
- (9) FIG. 5a is a view similar to FIG. 4a, wherein a guidewire is extended from the primary blood vessel and into the adjacent secondary blood vessel;

- (10) FIG. **5b** is a view similar to FIG. **4b**, wherein a guidewire is extended from the primary blood vessel and into the adjacent secondary blood vessel;
- (11) FIG. **6** illustrates the small communicating aperture and the guidewire placement created by the device and methods of the present invention after either embodiment of the inventive device of FIGS. **1a-5b** has been withdrawn from the procedural site;
- (12) FIG. **7** illustrates an isolated detail view of the distal tip of the piercing element for the illustrated embodiments;
- (13) FIG. **8** is a view similar to FIG. **2a**, illustrating a modified embodiment wherein a magnetic guidance system is deployed on each of the distal tip and a guidewire placed in the secondary vessel, for bringing the vessel walls together; and
- (14) FIG. **9** is a view similar to FIG. **8**, wherein the piercing element is advanced from the primary blood vessel into the adjacent secondary blood vessel, to thereby dislodge the magnetic guidewire.

DESCRIPTION OF THE PREFERRED EMBODIMENT

- (15) Referring now more particularly to the drawings shown in FIGS. **1a-7**, there are illustrated several embodiments of a device and system constructed in accordance with the principles of the present invention. As illustrated in FIG. **1a**, one embodiment of the device **10** comprises a handle or handpiece **2** and a main body shaft **12** having a secondary lumen **18** and a primary lumen **14** (FIG. **2a**). To begin the inventive method of intravascular access and communication, the practitioner selects an appropriate procedural site having each of a primary blood vessel **24** and a secondary blood vessel **26** (FIG. **1**) in close proximity to one another. In currently preferred approaches, the primary blood vessel **24** comprises a vein, and the secondary blood vessel **26** comprises an artery, but the invention is not limited to this arrangement. The main body **12** is inserted into primary vessel **24** so that the distal end **32** thereof (FIG. **2a**) lies within the blood flow passage of the primary vessel. Preferably, this insertion step is performed using percutaneous technique, but open surgery may also be employed.
- (16) With reference now to FIG. **2a**, a piercing element **20** comprises a piercing element shaft **34**, lumen **22**, and a distal tip **36**, and can be adjustably oriented axially within the secondary lumen **18** of a needle guide **16**, and lumen **22** provides an externally communicating passage. A distal end **40** of the needle guide **16** comprises a blunt large diameter atraumatic tip, comprised of a polymer material, having a rounded distal edge. This distal end **40** also has features that make it visible under different imaging techniques, such as ultrasound, fluoroscopy, CT, or MRI. There is a coil **42** constructed of a radiopaque material, embedded in the polymer distal end **40**. Small particles of air or other radiopaque materials known to those skilled in the art may also be used to increase the radiopacity of the end **40**.
- (17) Referring to FIGS. **2a** and **3a**, the blunt distal end **40** is manipulated to contact an inner wall of the primary vessel and to push it into desired engagement with the adjacent wall of the secondary vessel, as shown in FIG. **3a**. The position of desired engagement is arranged to optimize the piercing step to be next described. The distal tip **36** of the piercing element **20** may be longitudinally extended with respect to the needle guide **16**, using a slide **8** on the handle **2**. A range of the radius of curvature may be imparted on the piercing element **20** by axially adjusting the position of the main body **12** relative to needle guide **16**, using a knob **4** on the handle **2**. A first, or straightened, position is illustrated in FIG. **2a**, where the distal tip **36** is within the secondary lumen **18** of needle guide **16**. As will be described more fully below, the retracted orientation is utilized during the initial device insertion steps, as well as the device withdrawal steps, while variable extended orientations are the operative orientation for creating the communication passageway and guidewire placement. Needle guide **16** is fabricated of a material that has shape memory properties that allow it to be held in an essentially axial position indefinitely by main body shaft **12**, while in the orientation shown in FIG. **2a**, and can achieve an incremental increase in the radius of curvature as main body shaft **12** is retracted, as shown in FIG. **3a**. This variable orientation of the radius of curvature may be desirable by the practitioner to more effectively aim the distal tip **36** of the

piercing element **20** in order to achieve a more desirable orientation for access from primary vessel **24** to secondary vessel **26**. In one version of this embodiment, the needle guide **16** is fabricated of a superelastic material, such as Nitinol, to achieve this curvature effect. In another version of the embodiment, the piercing element shaft **34** can be formed with a radius of curvature. The strength of the piercing element shaft **34** is such that as the main body shaft **12** is retracted the piercing element shaft **34** imparts the radius of curvature onto the needle guide **16**. However, it should be noted that the needle guide **16** need not necessarily be made of a superelastic material for this embodiment to function. Since the shape of the needle guide comes from the piercing element shaft **34**, its shape is determined by moving the primary lumen **14** axially.

(18) Referring again to FIGS. **1a** and **3a**, once the main body **12** is inserted into primary vessel **24** and advanced to the desired site determined by the practitioner using ultrasound or fluoroscopic imaging, as previously described, it may be desired to adjust the radius of curvature of needle guide **16** to increase the angle of the axis of distal tip **36** by rotating knob **4** of handle **2**. Since piercing distal tip **36** is configured to have echogenic and radiopaque properties to allow the practitioner to visualize the orientation of piercing tip **36** under real time imaging guidance, and the main body **12** of device **10** is incrementally rotatable about its axis, this will allow the practitioner to more effectively aim piercing tip **36** through direct visualization as secondary blood vessel **26** is “nudged” by the atraumatic tip of the needle guide **16** of the device **10** as the main body is incrementally rotated and the radius of curvature as desired, to allow more accurate penetration from primary blood vessel **24** to secondary blood vessel **26**.

(19) With reference now to FIGS. **1a** and **4a**, once the practitioner has oriented piercing tip **36** as desired for optimal penetration, knob **8** of handle **2** is advanced to penetrate from primary blood vessel **24** through the primary vessel wall **44** to secondary blood vessel **26** through the secondary vessel wall **46**. This may be done under direct imaging guidance to verify complete penetration without extending beyond the flow passage of blood vessel **26**. The practitioner may also verify acceptable penetration through direct visualization of blood that flows through lumen **22** and exits through an aperture **6** of the handle **2** as shown in FIG. **1a**.

(20) With reference now to FIGS. **1a** and **5a**, once penetration from primary blood vessel **24** to secondary blood vessel **26** has been achieved, a guidewire **28**, preferably having a diameter of 0.014” or less, is advanced through the aperture **6** of the handle **2** until the guidewire is positioned in the blood flow path of blood vessel **26** sufficiently to allow device **10** to be removed while retaining its position in blood vessel **26**.

(21) With reference now to FIG. **6**, once guidewire **28** is sufficiently in position as previously described, the practitioner withdraws the device **10** completely from the body, thus leaving the guidewire in the desired position and crossing from primary vessel **24** to secondary vessel **26**.

(22) FIG. **7** illustrates a detail view of the configuration of the piercing tip **36** utilized in both of the illustrated embodiments. The tip is configured to have a lancet point **48** to enhance the penetration from primary blood vessel **24** to secondary blood vessel **26**. A primary bevel **50** is ground at an angle between 12 and 20 degrees with a secondary angle between 5-20 degrees, with a rotation angle between 25-45 degrees. The needle grind is designed such that it pierces through the vessel wall and does not core, or cut a plug, through the vessel wall, to minimize bleeding between vessels when removed after the guidewire is placed into the secondary vessel. The outer diameter of the piercing member is also minimized to further reduce bleeding. The piercing member is oriented within the secondary lumen such that the tip of the lancet point is directed toward the adjacent secondary vessel. Other piercing mechanisms, or needle point grind configurations, known to those skilled in the art may be provided.

(23) The embodiment of FIGS. **1b**, **2b**, **3b**, **4b**, and **5b** (the “B” embodiment) is similar in most respects to that of FIGS. **1a**, **2a**, **3a**, **4a**, and **5a** (the “A” embodiment), differing only in the details to be explained below. All common elements to those in the A embodiment are identified by common reference numerals in the figures illustrating the B embodiment, and the method

sequencing shown in FIGS. **2b**, **3b**, **4b**, and **5b** is similar to that shown in FIGS. **2a**, **3a**, **4a**, and **5a**. FIGS. **6** and **7** are common to both embodiments.

(24) The major difference between the A and B embodiments is that in the B embodiment the primary lumen **14** has been eliminated. This is because, in this embodiment, the shape of the needle guide **16** is not adjustable. Thus, it remains straight, and need not be fabricated of superelastic material. This arrangement is possible because the blunt distal end **40** may be manipulated by the practitioner to ensure that the adjacent vessel walls of the primary and secondary vessel may be pierced by an axial advancement of the piercing member, as shown in FIG. **3b**. As a result of this change, the knob **4** has also been eliminated, since control of the curvature of needle guide **16** is not required.

(25) Now with reference to FIGS. **8** and **9**, an additional embodiment employing innovative magnetic guidance systems and techniques will be described. In this embodiment, like components and structures to those in prior embodiments are denoted by like reference numerals. Notably, in this embodiment, the blunt distal end **40** of the needle guide **16** includes an alignment attachment, implant, or multiple implants embedded in or attached to the distal alignment end **40**. In this embodiment, a guidewire **52** is placed into the secondary blood vessel **26**, as shown, wherein the guidewire **52** has an alignment tip **54**. In the illustrated embodiment, the alignment end **40** and alignment tip **54** each comprise a magnetic material, either entirely or as an implant or multiple implants embedded in or attached to the each respective end **40** and tip **54**. The alignment tips **40** and **54**, whether comprised of permanent magnetic material or electromagnets, are established with opposite polarity, so that they are mutually attracted to one another.

(26) Each of the alignment members **40** and **54** discussed above, may comprise, in addition to magnetic attachments or implants, proximity sensors or ultrasonic sensors, as well as any other equivalent devices or systems for facilitating alignment of two members separated by opaque tissue, where imaging alignment procedures are less effective.

(27) Thus, as with the previous embodiments, to begin the inventive method of intravascular access and communication, the practitioner selects an appropriate procedural site having each of a primary blood vessel **24** and a secondary blood vessel **26** in close proximity to one another. In currently preferred approaches, the primary blood vessel **24** comprises a vein, and the secondary blood vessel **26** comprises an artery, but the invention is not limited to this arrangement. The main body **12** is inserted into primary vessel **24** so that the distal end **32** thereof (FIG. **2a**) lies within the blood flow passage of the primary vessel. Preferably, this insertion step is performed using percutaneous technique, but open surgery may also be employed.

(28) Referring to FIG. **8**, once the piercing element **20** is adjustably oriented axially within the secondary lumen of a needle guide, and these elements are further adjustably oriented axially within lumen **18** of the needle guide **16** (see FIG. **2a**), the lumen **22** provides an externally communicating passage. The distal end **40** of the needle guide **16**, as noted above, comprises a magnetic material.

(29) The magnetic distal end **40** is manipulated to a position proximate to or in contact with an inner wall of the primary vessel, as shown in FIG. **8**, at a location desirable for the creation of an AVF. Contemporaneously, the guidewire **52**, with magnetic tip **54**, is maneuvered within the blood vessel **26** to the same location. At this juncture the magnetic tips **40** and **54**, once they are maneuvered to locations adequately proximate to one another, become magnetically attracted to one another through the tissue walls of the respective vessels **24**, **26**. This mutual magnetic attraction causes the tips **40** and **54** to approach one another and to come into alignment, thus also functioning to physically push the vessels **24** and **26** together and into alignment as well. This alignment is shown in FIG. **8**. The respective magnetic tips provide good tactile and visual feedback to the practitioner when they are engaged, permitting confidence in knowing that the vessels **24** and **26** are aligned. The alignment of the vessels **24** and **26** optimizes the piercing step to be next described. The distal tip **36** of the piercing element **20** may be longitudinally extended with

respect to the needle guide **16**, between a range of the radius of curvature along the axis of needle guide **16**, using a slide **8** on the handle **2**. A first, or retracted, position is illustrated in FIG. **8**. However, in FIG. **9**, the distal tip **36** of the needle or piercing element **20** has been extended beyond the end of needle guide **16**, and through the adjacent tissue walls of each vessel **24**, **26**.

(30) Once penetration from primary blood vessel **24** to secondary blood vessel **26** has been achieved, thereby creating a small communicating aperture between the two vessels, the needle functions to dislodge the arterial magnet **54** by pushing and advancing the magnet **54** away from the magnetic tip **40** and advances its attached guidewire **52** as well. A guidewire **28** may be disposed through the procedural site and device **10** may be withdrawn, as shown in FIG. **6** and as discussed in connection with the prior embodiment. Instrumentation may then be introduced to the procedural site over the guidewire **28** to create the desired AVF, as taught and disclosed, for example, in commonly assigned U.S. Pat. Nos. 9,439,710, 9,452,015, and 9,474,562, each expressly incorporated herein by reference.

(31) Accordingly, although an exemplary embodiment and method according to the invention have been shown and described, it is to be understood that all the terms used herein are descriptive rather than limiting, and that many changes, modifications, and substitutions may be made by one having ordinary skill in the art without departing from the spirit and scope of the invention.

Claims

1. A method of creating intravascular access, comprising: manipulating a distal end of a device having a first alignment member to a location proximate to an inner wall of a primary vessel; manipulating a guidewire having a second alignment member to a location proximate to an inner wall of a secondary vessel; magnetically coupling the first alignment member of the distal end of the device and the second alignment member of the guidewire through the respective walls of the primary and secondary vessels, so that the device and guidewire are in close proximity and in alignment, and thereby pushing the primary and secondary vessels together; and extending, after the step of magnetically coupling, a piercing member distally from the device, through the wall of the primary vessel, and through an adjacent wall of the secondary vessel, so that the end of the piercing member is disposed within the secondary vessel and thereby creating a communicating aperture on the opposing walls of the primary and secondary vessel, contacting, during the step of extending the piercing member, the second alignment member with the piercing member to uncouple the magnetically coupled first and second alignment members and move both the second alignment member and the guidewire away from the distal end of the device.
2. The method as recited in claim 1, further comprising inserting the device into the primary vessel, wherein the inserting step is performed percutaneously.
3. The method as recited in claim 1, and further comprising a step of withdrawing the device from the primary vessel.
4. The method as recited in claim 1, wherein the first alignment member on the device is comprised of one or more of a magnetic attachment, a magnetic implant, a proximity sensor, and an ultrasonic sensor.
5. The method as recited in claim 1, wherein the second alignment member comprises one or more of a magnetic attachment, a magnetic implant, a proximity sensor, and an ultrasonic sensor.
6. The method as recited in claim 1, wherein the guidewire has a guidewire body, and the second alignment member comprises a magnetic material and projects laterally outward from a lateral-most dimension of the guidewire body.
7. The method as recited in claim 6, wherein the magnetic material extends laterally from the guidewire body in a first direction and in a second direction opposite the first direction.
8. The method as recited in claim 1, wherein the guidewire has a guidewire body, the second alignment member having a blunt distal surface extending laterally outward from the guidewire

body, the blunt distal surface being configured to move the secondary vessel toward the primary vessel.

9. The method as recited in claim 1, wherein the guidewire has a guidewire body, an entirety of the second alignment member being disposed distally of the guidewire body.
 10. The method as recited in claim 1, wherein the second alignment member comprises a continuous block of magnetic material.
 11. The method as recited in claim 1, wherein the second alignment member defines a distal surface extending continuously across a distal-most end of the second alignment member.
 12. The method as recited in claim 1, wherein the primary vessel comprises a vein, and the secondary vessel comprises an artery.
 13. The method as recited in claim 12, further comprising creating an arterio-venous fistula (AVF) by connecting the artery to the vein.
 14. The method as recited in claim 1, wherein each of first and second alignment members comprises magnetic material.
 15. The method as recited in claim 14, wherein the magnetic material is permanent magnetic material.
 16. The method as recited in claim 14, wherein at least one of the first and second alignment members further comprises at least one of a proximity detector or an ultrasonic sensor.
 17. The method as recited in claim 1, further comprising: advancing, after the step of contacting the second alignment member with the piercing member to uncouple the magnetically coupled first and second alignment members, a second guidewire through a lumen of the piercing member so that the guidewire enters the secondary vessel.
 18. The method as recited in claim 17, further comprising: withdrawing, after the step of advancing the second guidewire through the lumen of the piercing member, the device from the primary vessel.
 19. The method as recited in claim 18, wherein after said withdrawing the device from the primary vessel, the second guidewire extends from the primary vessel, through the communicating aperture, and into the secondary vessel.
 20. The method recited in claim 19, wherein the primary vessel comprises a vein, and the secondary vessel comprises an artery, wherein the method further comprises: advancing, after said withdrawing the device from the primary vessel and with the second guidewire extending from the primary vessel, through the communicating aperture, and into the secondary vessel, instrumentation over the guidewire to the communicating aperture; and creating an arterio-venous fistula (AVF) by connecting the artery to the vein using the instrumentation.
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